

Sakshi Suryawanshi

Full Stack Software Engineer

sakshi.suryawanshi520@gmail.com | India | Open to global remote opportunities

[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Professional Profile

Full Stack Software Engineer with 2+ years of production experience across backend systems, REST APIs, analytical data platforms, and cloud infrastructure. Currently contributing to a multi-tenant asset performance management SaaS at **Eightgen AI**, building **Python/FastAPI** microservices, **ClickHouse**-backed analytics, and an AI conversational analytics service on **LangGraph + Google Vertex AI Gemini**. Previously at **Wassha Inc. (Japan)**, independently owned and shipped a supply chain management platform end-to-end, mentored 6 engineers, and delivered measurable improvements in system performance, deployment speed, and reliability. Experience spans responsive web applications, real-time features with WebSockets, full-stack performance optimisation, automated CI/CD, and AWS infrastructure management. Comfortable in **AI-augmented development workflows** (Claude Code, Cursor, GitHub Copilot), async-first remote teams, and international collaboration across time zones, with a proven ability to drive architectural improvements and produce clear technical documentation.

Key Achievements

- **Multi-Tenant Security Hardening** — Closed a critical cross-tenant exposure path in Eightgen's APS scoring service by enforcing `client_id` isolation across every ClickHouse analytical query
- **AI Conversational Analytics (in design)** — Architecting a Text-to-SQL microservice on LangGraph + Google Vertex AI Gemini with a 9-layer SQL safety stack (AST validation, RBAC, tenant isolation, deny-by-default allowlists)
- **40% Performance Improvement** — Optimised full-stack systems through query optimisation, legacy code refactoring, and frontend performance enhancements across production endpoints
- **3x Reduction in Production Incidents** — Proposed and implemented AWS Application Load Balancer enhancements on an inherited legacy monolith, transforming service reliability
- **20% Faster Deployments** — Designed and implemented CI/CD pipelines using Docker and Kubernetes, automating build and deployment workflows end to end
- **6 Engineers Mentored** — Led structured onboarding, regular code reviews, and pair programming sessions; cut average ramp-up time by 50%
- **Real-time at Scale** — Built WebSocket and Socket.io-based communication systems delivering sub-second notifications to thousands of concurrent users
- **95% ML Model Accuracy** — Developed speech emotion recognition system on 8,000+ audio samples; published research as a peer-reviewed book chapter
- **9.54/10 CGPA** — Graduated with Honours in Data Science from Computer Science Engineering program

Professional Experience

Software Engineer (Contract)

Mar 2026 – Present

Eightgen AI | Remote

Asset Performance Management SaaS — manufacturing industry client

Stack: Python, FastAPI, Pydantic, SQLAlchemy, asyncpg, PostgreSQL, ClickHouse, React 19, TypeScript, TanStack, pytest-bdd, GitHub Actions, LangGraph, Google Vertex AI Gemini, sqlglot

Backend & Data Platform

- Hardened **multi-tenant data isolation** across the APS execution engine by injecting `client_id` filters throughout every **ClickHouse** analytical query, and resolved ClickHouse-specific correctness issues (FINAL-keyword deduplication, JSONB type coercion for asyncpg) — closing a critical cross-tenant exposure path in the scoring service
- Architected the end-to-end **benchmark lifecycle** with **copy-on-write versioning** (clone → DRAFT → ACTIVE), status transitions, default initialization, and administrative overrides on a multi-tenant PostgreSQL schema, enabling safe iteration on scoring rules without disrupting live benchmarks
- Eliminated **N+1 queries** in the scoring data transformation pipeline via a bulk-UPDATE strategy over JSONB-embedded quarterly period scores, and rewrote `list_results` as a **single-pass query** with a composite index, materially reducing dashboard latency under multi-tenant load

Full-Stack Dashboards & Quality

- Delivered interactive **score-trend visualizations** and a historical-results browser (filters, pagination, insight-name resolution) in **React 19 / TypeScript / TanStack**, backed by ClickHouse analytics served through FastAPI, plus a UX guard dialog preventing invalid analysis periods on the APS execute page
- Authored **pytest-bdd** end-to-end scenario coverage across RBAC, tenant isolation, and calculation correctness; produced Technical Approach Documents for benchmark management, score-trend charts, and scenario gap analysis
- Restored reliability in **GitHub Actions CI/CD** deploy workflows by injecting missing service URL and webhook environment variables; narrowed exception handling in admin invitation flows

AI / LLM Engineering (in progress)

- Designing a **multi-tenant AI Conversational Analytics microservice** on **LangGraph** StateGraph agents running against **Google Vertex AI Gemini** (Flash Lite / Flash / Pro routing), with a Text-to-SQL pipeline secured by a **9-layer safety stack** — sqlglot AST validation, deny-by-default allowlists, RBAC column authorization, and tenant filter injection
- Specifying **conversation persistence** (PostgreSQL checkpoint for LangGraph state), **SSE streaming** response delivery, and a full query audit log for compliance and drill-downs

Full Stack Software Engineer

Mar 2024 – Jan 2026

Wassha Inc., Japan | *Remote (software department fully remote)* *Energy-access technology, operations across East Africa*

Stack: Python, Django, ReactJS, TypeScript, NestJS, Node.js, MySQL, PostgreSQL, Redis, AWS (EC2, RDS, S3, Redshift, Glue), Docker, Kubernetes, Pulumi, WebSockets, Socket.io, CI/CD, Nginx, Gunicorn, CloudWatch, Grafana Loki, Apache Superset, Android Studio

Full Stack System Ownership & Delivery

- **Independently owned and delivered** a full-stack supply chain management platform, overseeing the complete development lifecycle including frontend UI/UX, backend APIs, database design, deployment, monitoring, and post-release support
- Contributed to **3 internal production projects concurrently**, demonstrating adaptability across frontend and backend codebases while maintaining high code quality and system stability
- Collaborated with cross-functional teams to deliver **business-critical features** consumed internally and by customer-facing mobile applications serving **thousands of users**

Backend Architecture & API Development

- **Independently owned** the full backend lifecycle of the supply chain platform, covering API design using **Django REST Framework** and **NestJS**, database schema, deployment, and post-release monitoring
- Designed and developed **RESTful APIs** with Django REST Framework and NestJS, ensuring proper authentication, validation, and error handling throughout
- Optimised **MySQL and PostgreSQL** schemas with strategic indexing and eliminated redundant queries across high-traffic reporting endpoints, contributing to a **40% improvement** in overall system response time
- Applied strategic **database indexing and query optimisation** techniques to significantly improve response times for high-traffic endpoints and reporting modules
- Proposed and implemented **AWS Application Load Balancer** enhancements on an inherited legacy monolith, reducing production incident frequency by **over 3x** and significantly improving service reliability
- Designed and implemented **Redis caching** strategies to reduce database load and improve API response times across frequently accessed endpoints
- Proposed and led the **migration of two repositories from Django to NestJS**; integrated multiple third-party APIs, improving system modularity and long-term maintainability
- Assumed **full ownership** of a critical production data pipeline incident where a schema change caused Redshift ingestion failures and broken analytics dashboards; diagnosed root cause, restored data consistency, fixed affected AWS Glue jobs, and authored a **post-mortem** to prevent recurrence
- Architected **dynamic country-aware credential handling** logic, replacing hardcoded configurations with a scalable, maintainable solution deployed across multiple regions
- Contributed to **unit and integration testing** across backend services using pytest; maintained **Swagger/OpenAPI** documentation for all API endpoints

Frontend Architecture & Performance

- Built reusable **ReactJS and TypeScript component library** (dynamic forms, data tables, interactive

- dashboards) with code splitting and lazy loading, reducing frontend development time across 3 concurrent projects
- Implemented **frontend performance optimisations** including code splitting, lazy loading, and efficient state management, contributing to the overall 40% system performance improvement
 - Optimised **JavaScript execution and DOM manipulation** to reduce page load times and improve user experience across the platform
 - Built reusable component libraries following best practices for **maintainability and scalability**

Real-time Communication & Event-Driven Features

- Integrated **Socket.io** to deliver **sub-second real-time notifications** and live status updates for **thousands of concurrent users**, replacing a polling-based approach and reducing unnecessary server load
- Developed and integrated **WebSocket-based real-time communication** features for instant notifications, live status updates, and bidirectional event-driven communication
- Built **event-driven architecture** supporting real-time updates across multiple concurrent users and sessions
- Designed and implemented **connection management, reconnection logic, and error handling** for WebSocket communications to ensure reliability under variable network conditions

Database Design & Management

- Designed **relational database schemas** for MySQL and PostgreSQL, ensuring normalisation, referential integrity, and optimal query performance
- Implemented **Redis caching** strategies to reduce database load and improve API response times across frequently accessed endpoints
- Managed **data pipelines involving AWS Redshift** for analytics workloads, including schema migrations and pipeline maintenance

DevOps, Cloud & Reliability

- Designed and implemented **CI/CD pipelines** using **Docker and Kubernetes**, automating build and deployment workflows and reducing delivery time by **20%**
- Managed **AWS infrastructure** across EC2, RDS, S3, Redshift, and Glue for production applications, analytics workloads, and data pipelines
- Developed and maintained **Kubernetes deployment workflows** for batch processing jobs, ensuring reliable and repeatable deployment cycles
- Monitored system health and debugged production issues via **CloudWatch and Grafana Loki**, enhancing system observability and improving incident response times

Leadership & Mentorship

- **Mentored 6 junior engineers** through structured onboarding programs covering both frontend and backend development, regular code reviews, and collaborative pair programming — cutting average ramp-up time by **50%**
- Led complex full-stack initiatives including **multiple third-party API integrations** and strategic repository migration from Django to NestJS
- Authored and maintained **comprehensive developer documentation** for frontend components, backend APIs, and deployment runbooks, improving onboarding efficiency and ensuring long-term system maintainability
- Conducted multiple **internal knowledge transfer sessions** on full-stack development workflows, React best practices, and production debugging
- Contributed to **sprint planning and backlog prioritisation** in an async-first remote team, effectively balancing business-critical features with technical maintenance

Mobile App Testing

- Conducted **manual testing of mobile applications** using Android Studio, identifying and reporting UI/UX issues and functional bugs
- Collaborated with mobile development teams to ensure **consistent user experience** across web and Android platforms

Recommendation Letter from Head of Software Engineering | *Experience Letter* issued by Wassha Inc., Japan

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL

Backend & APIs: FastAPI, Pydantic, SQLAlchemy, async/await, asyncpg, Django, Django REST Framework, Flask, Node.js, Express.js, NestJS, REST APIs, GraphQL, Celery, WebSockets, Socket.io, SSE streaming

Databases: PostgreSQL, ClickHouse, MySQL, SQLite, MongoDB, Redis, AWS Redshift

Frontend: React (19), NextJS, TypeScript, TanStack, HTML5, CSS3, TailwindCSS, Bootstrap, ES6+, component architecture, state management

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda, Redshift, Glue), GCP (Cloud Run, Cloud SQL, GCS — familiarity), Docker, Kubernetes, Pulumi, GitHub Actions, CI/CD, Nginx, Gunicorn

AI / LLM Engineering: LangGraph, Google Vertex AI Gemini, OpenAI, Anthropic, Text-to-SQL, sqlglot, RAG patterns, prompt engineering, multi-layer SQL safety validation

AI-Augmented Development: Claude Code, Cursor, GitHub Copilot, AI-assisted code review and refactoring

Real-time Technologies: WebSockets, Socket.io, SSE, event-driven architecture, real-time data synchronisation

Testing & Observability: pytest, pytest-bdd, unit and integration testing, CloudWatch, Grafana Loki, Apache Superset, Swagger/OpenAPI, Git, GitHub, Jira, VS Code, Android Studio

Core Competencies: System design, microservices architecture, multi-tenant SaaS, data isolation & RBAC, async API design, query optimisation, technical documentation, Agile/Scrum, Data Structures and Algorithms, OOP, database schema design

Projects

Task Management System

GitHub

Personal Project | Python, Django REST Framework, MySQL, Redis, Celery, Docker, JWT, Swagger

- **Architected and developed** a scalable backend system for task and project management utilising Django REST Framework
- Implemented **JWT-based authentication with RBAC** for secure, role-aware user management across teams, projects, and task hierarchies
- Integrated **Celery and Redis** for async background processing including automated email notifications and scheduled tasks
- **Containerised the full application stack** (MySQL, Redis, Nginx, Gunicorn, Celery) using Docker Compose for reproducible local and production environments
- Created interactive **Swagger/OpenAPI documentation** covering all API endpoints with request/response schemas
- Designed **relational database schema** with optimised queries for tasks, projects, teams, notifications, and activity logs

Mingling Minds

GitHub

Personal Project | NextJS, TypeScript, GraphQL, Apollo Server, Prisma, PostgreSQL, AWS

- Developed a **full-stack social media platform** featuring real-time feeds, post creation, and image uploads
- Built a modern, responsive frontend using **NextJS and React with TypeScript** for type safety and component reusability
- Implemented complex UI components including **infinite scroll, image galleries, and interactive forms**
- Designed **GraphQL API with Apollo Server** for efficient data fetching and real-time updates
- Implemented secure authentication using **Google OAuth 2.0** and JWT-based API authorisation
- Automated **GraphQL query generation using Codegen**; deployed on AWS with production architecture
- Integrated **Prisma ORM** for type-safe database access and schema management with PostgreSQL

Education

Bachelor of Engineering (B.E.) in Computer Science
Pimpri Chinchwad College of Engineering, Pune

CGPA: 9.54 / 10
June 2018 – June 2022

Honours in Data Science
View Degree Certificate

June 2020 – June 2022

Relevant coursework: Data Structures and Algorithms, OOP, Database Systems, OS, Computer Networks, ML, Software Engineering

Certifications & Achievements

View All Certificates

- **Published research:** “Emotion Recognition Systems Based on Machine Learning Classifiers” in *21st Century In-*

novations in Management, Science and Technology, Education, and Social Sciences; achieved **95% classification accuracy** on **8,000+ audio samples** using a multi-layer CNN

- **Finalist** at Smart PCMC Hackathon (2020)
- **Membership Chairperson** — PCCOE CSI Chapter (2020 – 2021)
- Lyft Back-End Engineering Virtual Experience — Forage
- Introduction to Back-End Development — Meta
- Python and SQL Certifications — HackerRank (**5-Star Rating in Python**)

Additional Information

- Conducted multiple **internal knowledge transfer sessions** on full-stack development workflows, React best practices, and production debugging techniques
- Experienced in **fully remote, international team environments** across multiple time zones with async-first communication
- Strong **communication skills and ownership mentality**; comfortable taking end-to-end responsibility for features and systems
- Passionate about learning new technologies and staying current with modern backend, full-stack, and AI-augmented development practices